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Assignment_02_CO_01

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QUESTIONS

- Q1 ✓ Print a hollow diamond pattern — 5 marks
- Q2 ✓ Design a class to calculate parking charges — 5 marks
- Q3 ✓ Print a number pattern with consecutive values — 5 marks
- Q4 ✓ Check whether a number is a spy number — 5 marks
- Q5 ✓ Print a butterfly pattern — 5 marks

5/5 answered

Q1 Print a hollow diamond pattern — 5 marks

Use nested loops to print a diamond of five rows in the upper half and four rows in the lower half.

```
*
**
**
**
**
**
**
**
*
```

Your Code

```c
#include <stdio.h>

int main() {
    int i, j, n = 5;

    for (i = 1; i <= n; i++) {
        for (j = 1; j <= n - i; j++)
            printf(" ");

        for (j = 1; j <= 2 * i - 1; j++) {
            if (j == 1 || j == 2 * i - 1)
                printf("*");
            else
                printf(" ");
        }
        printf("\n");
    }

    for (i = n - 1; i >= 1; i--) {
        for (j = 1; j <= n - i; j++)
            printf(" ");

        for (j = 1; j <= 2 * i - 1; j++) {
            if (j == 1 || j == 2 * i - 1)
                printf("*");
            else
                printf(" ");
        }
        printf("\n");
    }

    return 0;
}
```

Input

stdin input

Output

```
    *
   * *
  *   *
 *     *
*       *
 *     *
  *   *
   * *
    *
```

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5/5 answered

Q2 Design a class to calculate parking charges

5 marks

Create a Parking class. Charge ₹30 for the first hour and ₹20 for every additional hour. Any fraction of an hour counts as a full hour.
TC_1: 2 hours and 15 minutes →Parking charge: Rs.70. TC_2: 0 hours and 30 minutes →Parking charge: Rs.30 .TC_3: 2 hours and 60 minutes → Invalid duration

Your Code

```cpp
#include <iostream>
using namespace std;

class Parking {
public:
    int hours, minutes;

    void calculate() {
        if (hours < 0 || minutes < 0 || minutes >= 60) {
            cout << "Invalid";
            return;
        }

        int totalMinutes = hours * 60 + minutes;
        int totalHours = (totalMinutes + 59) / 60;

        int charge;
        if (totalHours == 0)
            charge = 0;
        else
            charge = 30 + (totalHours - 1) * 20;

        cout << "Parking charge: Rs." << charge;
    }
};

int main() {
    Parking p;
    cin >> p.hours >> p.minutes;
    p.calculate();
    return 0;
}
```

Input

```
2 15
70
```

Output

```
Parking charge: Rs.70
```

Visible Test Cases

- Test Case 1 ✓
- Test Case 2 ✓
- Test Case 3 ✗

4.00 / 5 marks

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- Q5 ✓ Print a butterfly pattern — 5 marks

5/5 answered

Q3 Print a number pattern with consecutive values

5 marks

Print the following pattern for four rows.

1
2 3
4 5 6
7 8 9 10

Your Code

```cpp
#include <iostream>
using namespace std;

int main() {
    int n = 1;

    for (int i = 1; i <= 4; i++) {
        for (int j = 1; j <= i; j++) {
            cout << n;
            n++;
        }
        cout << endl;
    }

    return 0;
}
```

Input

stdin input

Output

```
1
23
456
78910
```

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- Q5 ✓ Print a butterfly pattern — 5 marks

5/5 answered

Q4 Check whether a number is a spy number — 5 marks

A number is a spy number if the sum of its digits equals the product of its digits.
TC_1:1124 → Spy number. TC_2:124 → Not a spy number. TC_3:-123 → Invalid input

Your Code

```cpp
#include <iostream>
using namespace std;

int main() {
    int n, digit, sum = 0, product = 1;
    cin >> n;

    if (n < 0) {
        cout << "Invalid input";
        return 0;
    }

    while (n > 0) {
        digit = n % 10;
        sum += digit;
        product *= digit;
        n /= 10;
    }

    if (sum == product)
        cout << "Spy number";
    else
        cout << "Not a spy number";

    return 0;
}
```

Input

```
1124
```

Output

```
Not a spy number
```

Visible Test Cases

- Test Case 1 ✓
- Test Case 2 ✓
- Test Case 3 ✓

4.00 / 5 marks

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- Q5 ✓ Print a butterfly pattern — 5 marks

5/5 answered

Q5 Print a butterfly pattern — 5 marks

Print a butterfly pattern using nested loops.

```
* *
** **
*** ***
**** ****
**********
**** ****
*** ***
** **
* *
```

Your Code

```cpp
#include <iostream>
using namespace std;

int main() {
    int n = 4;

    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= i; j++) {
            cout << "*";
        }
        cout << endl;
    }

    for (int i = n - 1; i >= 1; i--) {
        for (int j = 1; j <= i; j++) {
            cout << "*";
        }
        cout << endl;
    }

    return 0;
}
```

Input

stdin input

Output

```
*
**
***
****
***
**
```

Submitted

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